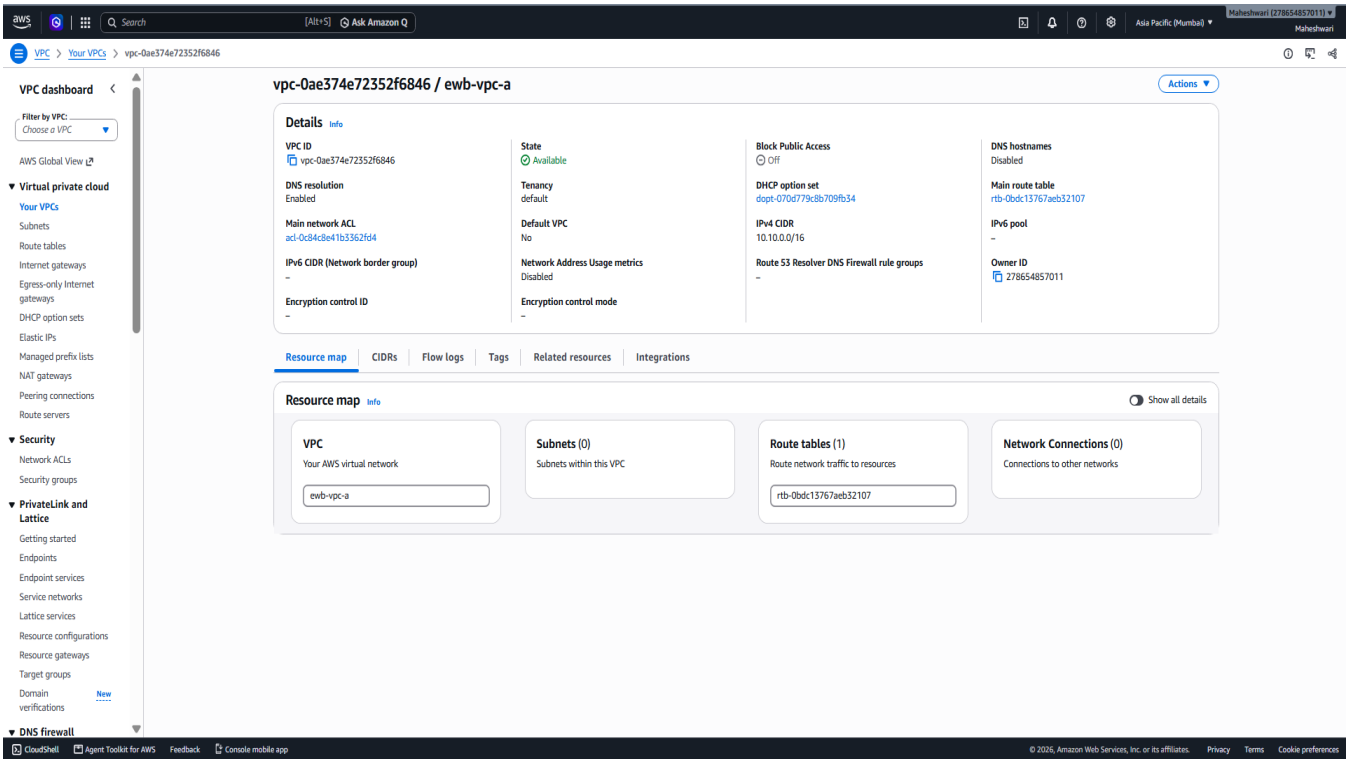
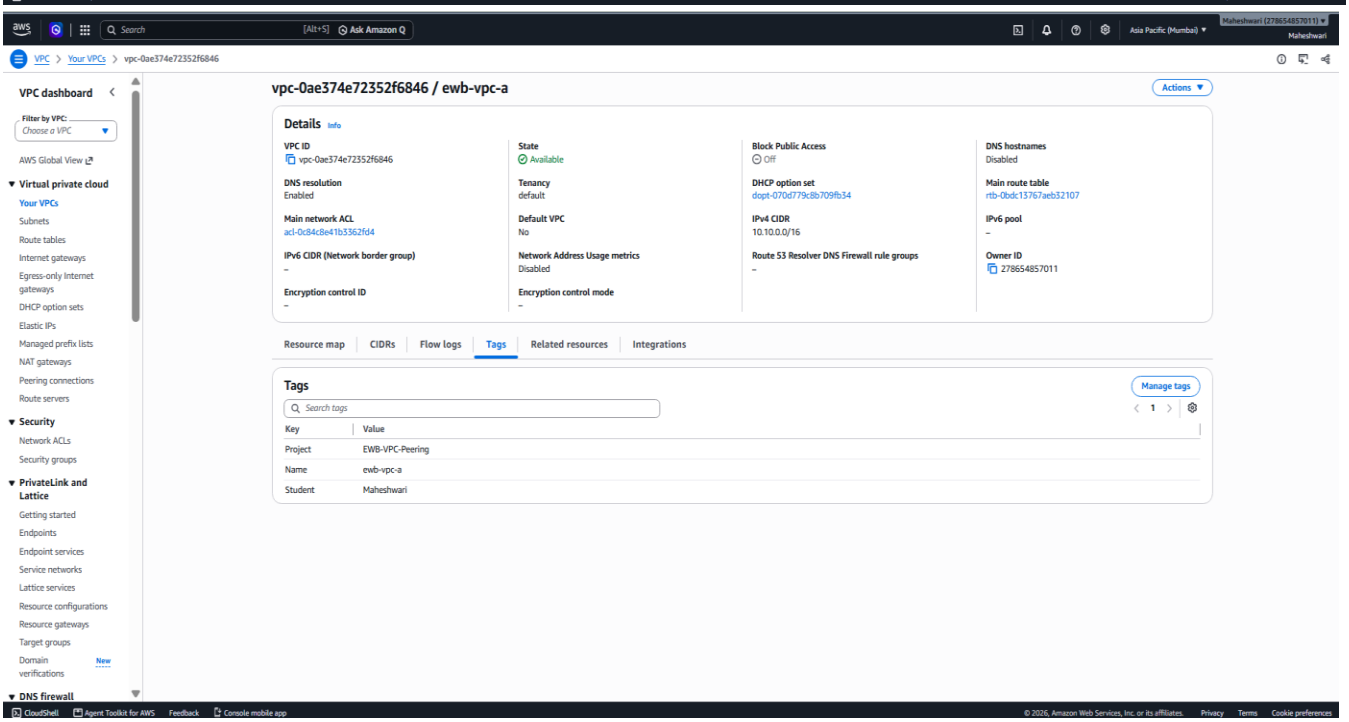
VPC Project

1. VPC-A

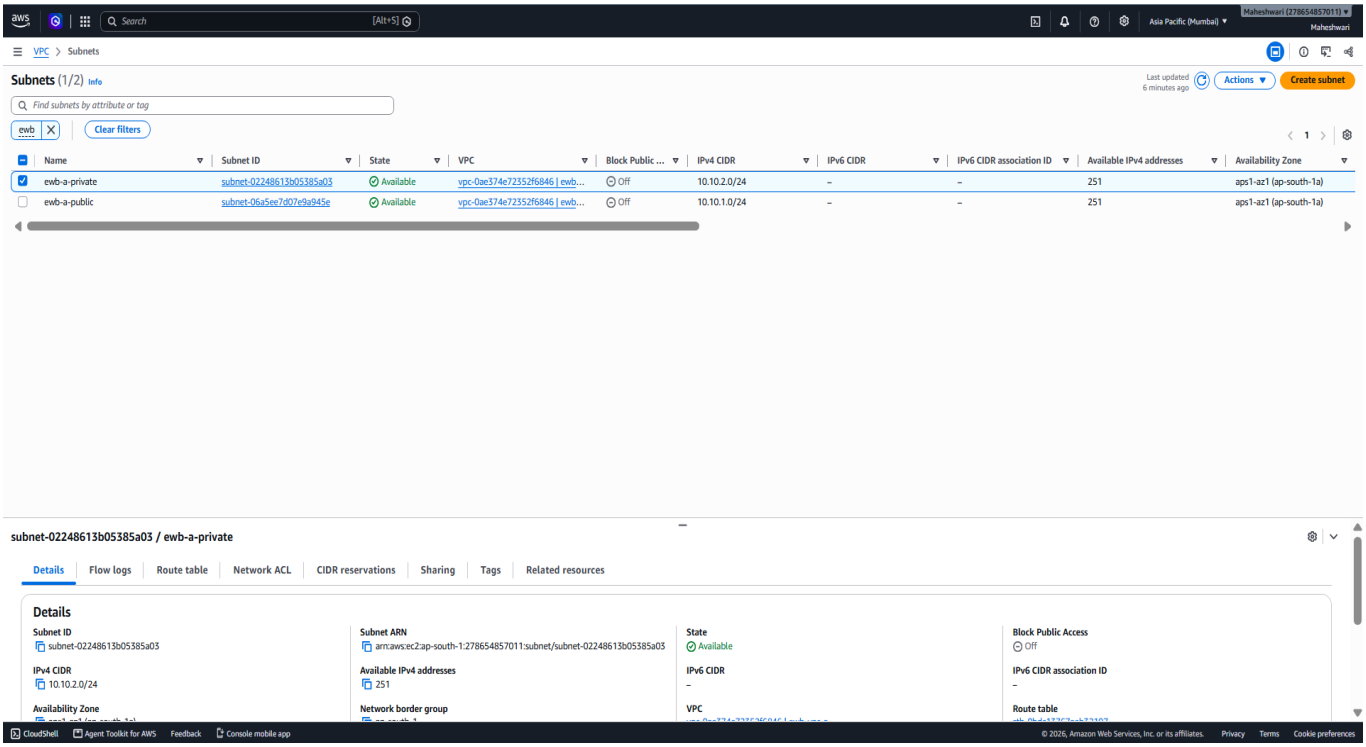
Created ewb-vpc-a (10.10.0.0/16) with the tags Project = EWB-VPC-Peering and Student = Maheshwari.





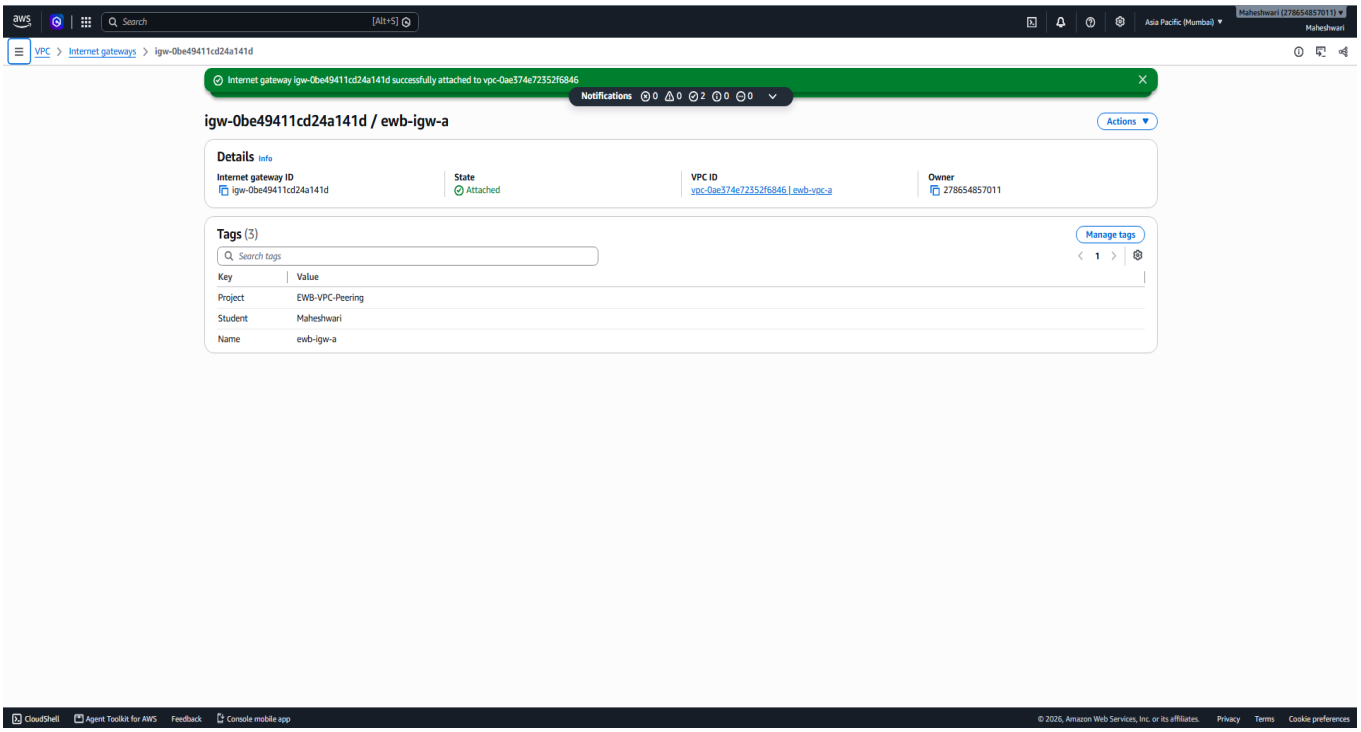
Step 2: Subnets in VPC A

Created two subnets in ap-south-1a: ewb-a-public (10.10.1.0/24) for the bastion and ewb-a-private (10.10.2.0/24) for App-A.

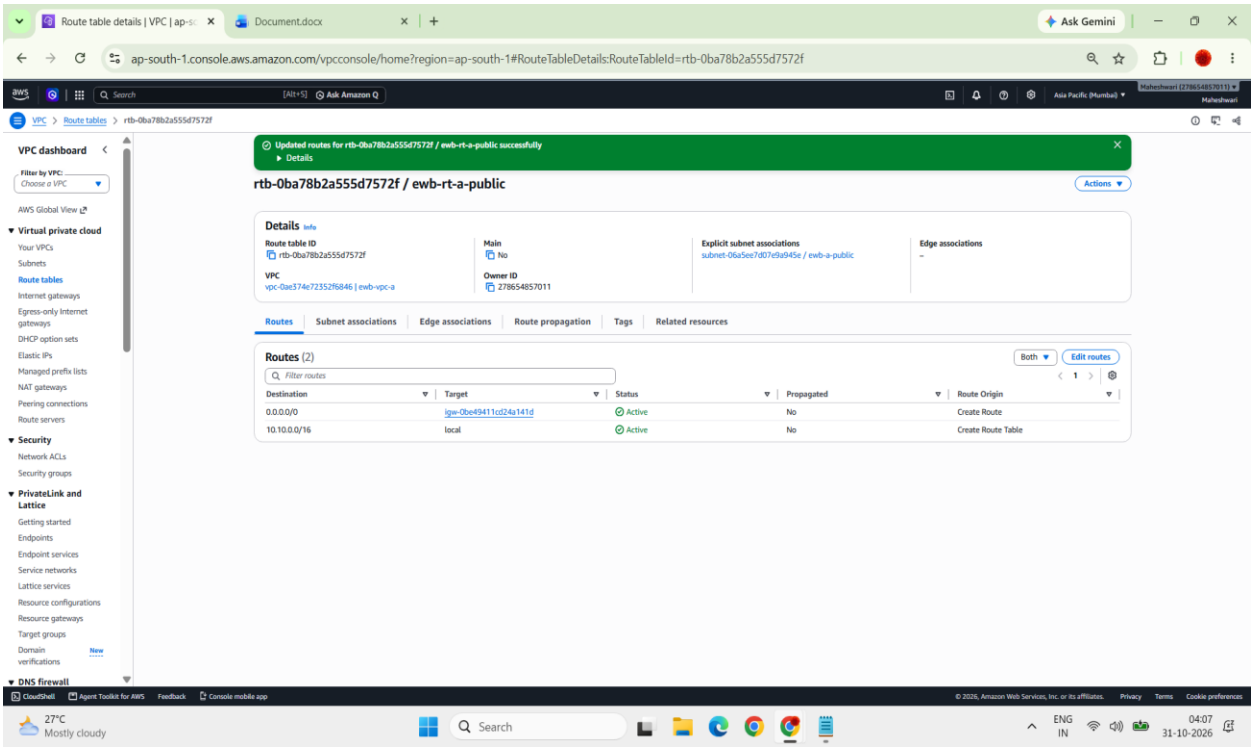


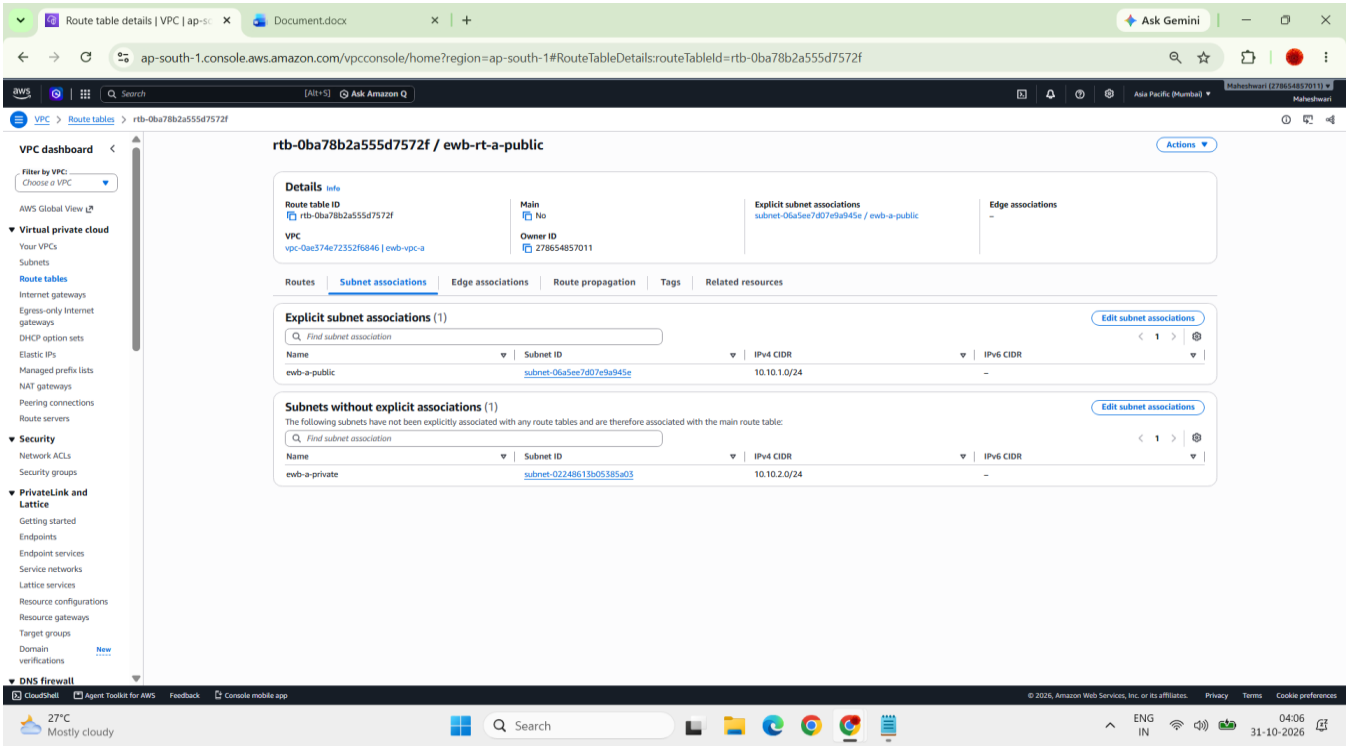
Step 3: Internet gateway

Created ewb-igw-a and attached it to ewb-vpc-a so that the public subnet can reach the internet.



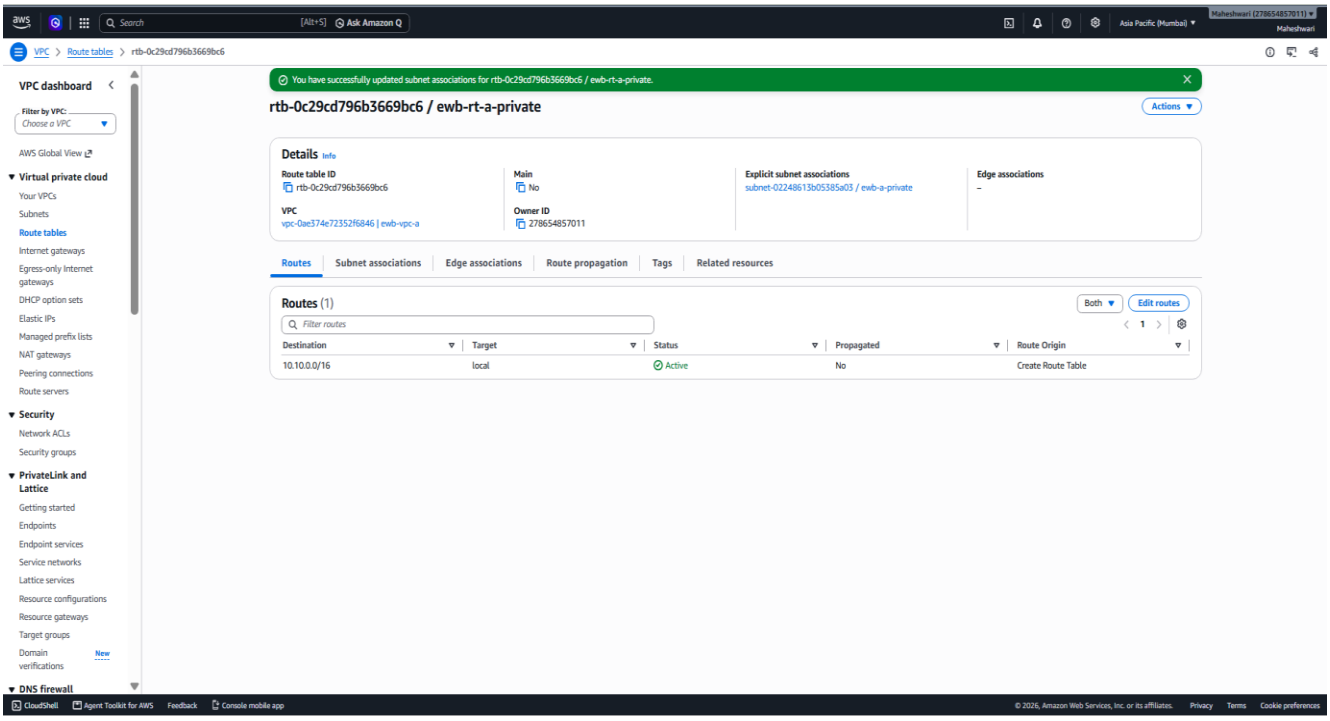
ewb-rt-a-public has the route 0.0.0.0/0 to ewb-igw-a and is associated with ewb-a-public. This makes that subnet public.

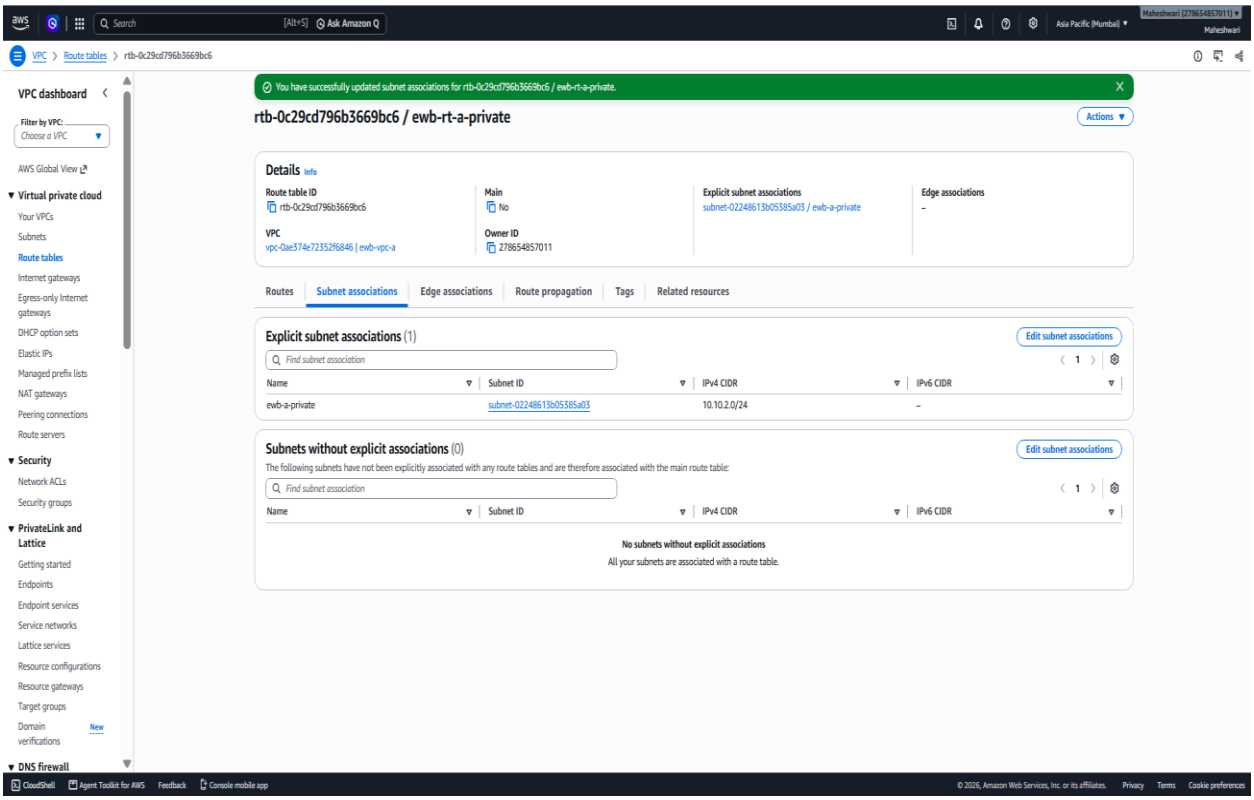




Step 5: Private route table

ewb-rt-a-private has only the local route (10.10.0.0/16) and is associated with ewb-a-private. It has no internet route, so the subnet is private.

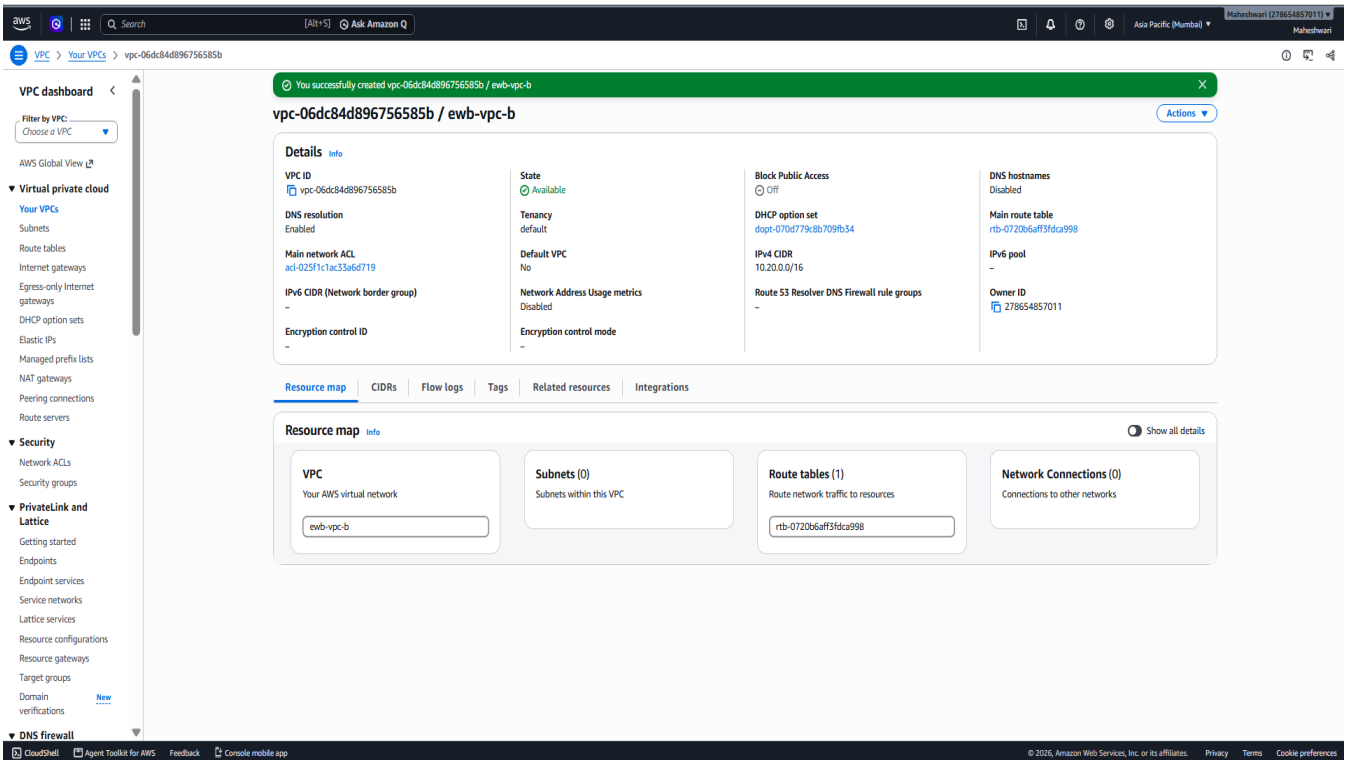


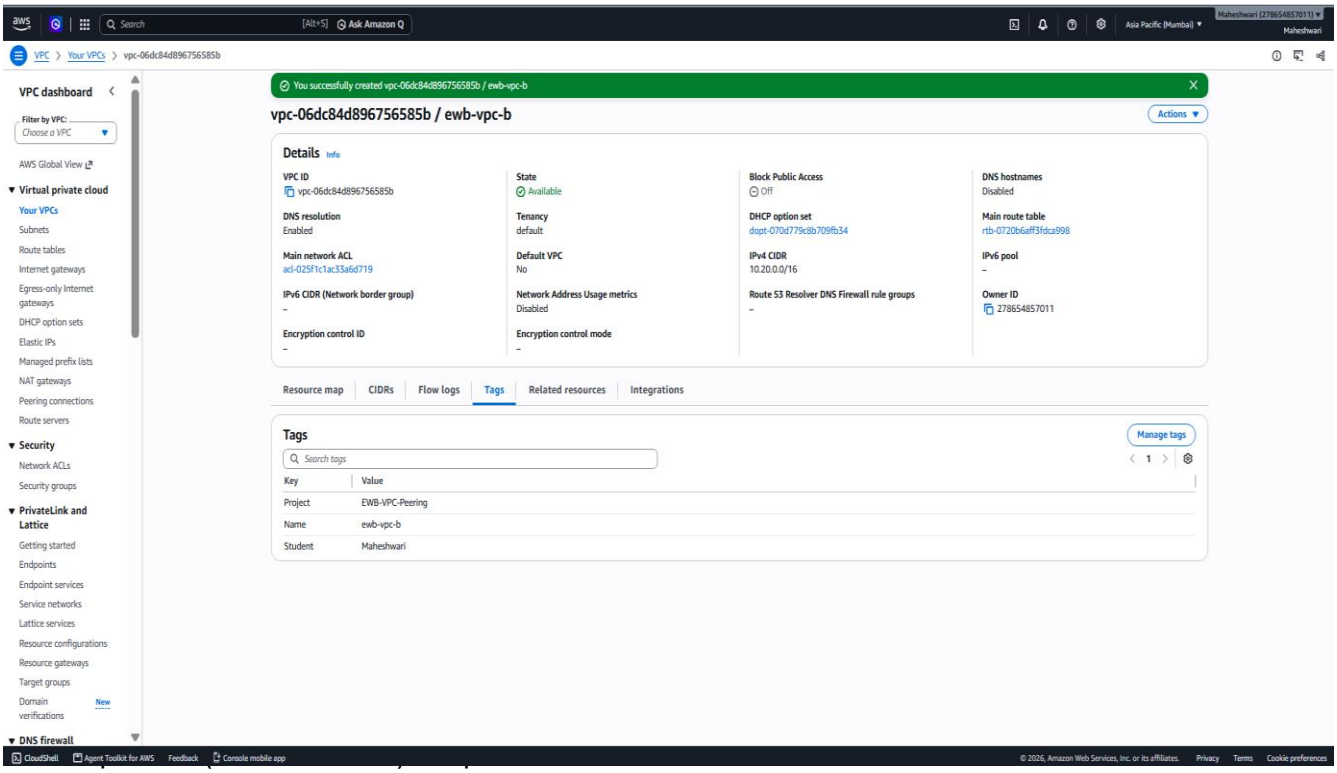


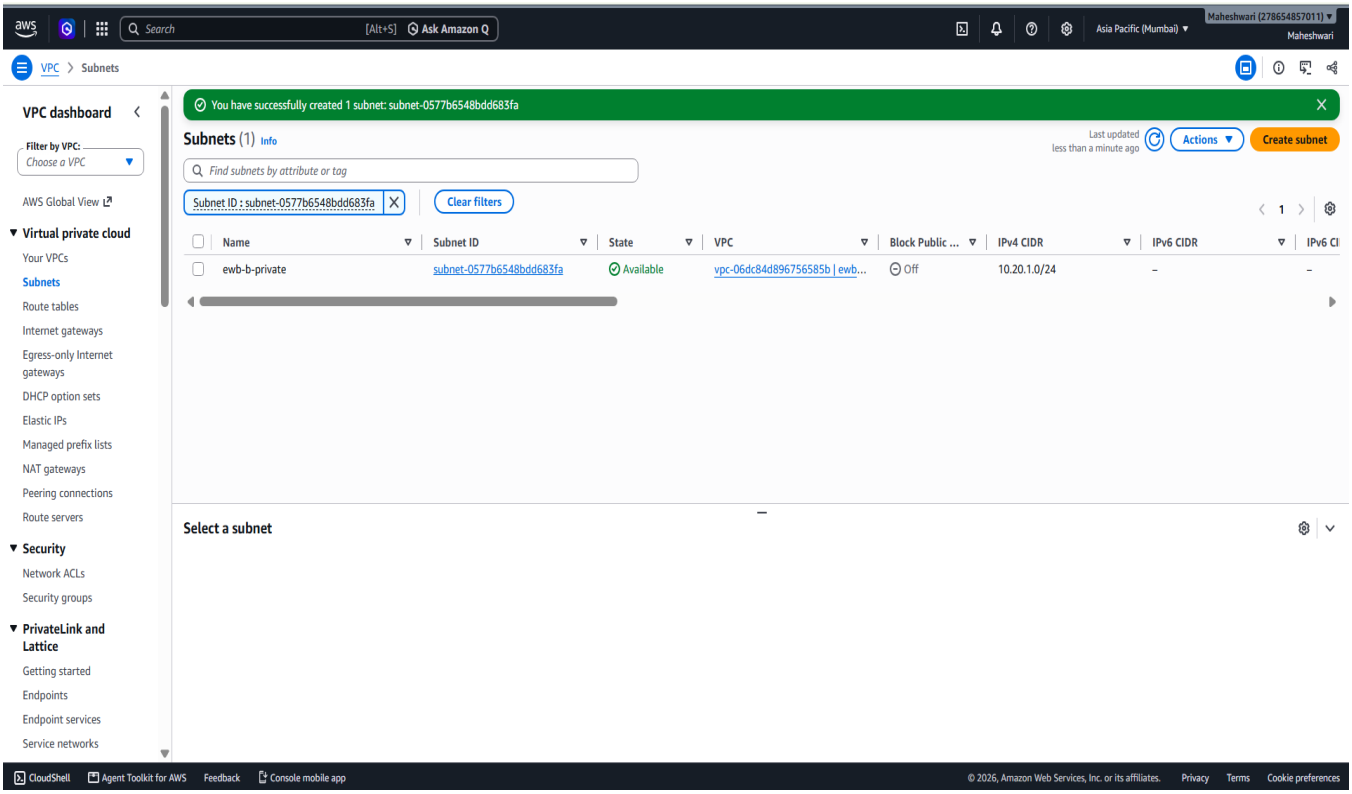
Step 6: VPC B

Created ewb-vpc-b (10.20.0.0/16). Its range does not overlap with VPC A, which VPC peering requires.

6a: VPC B



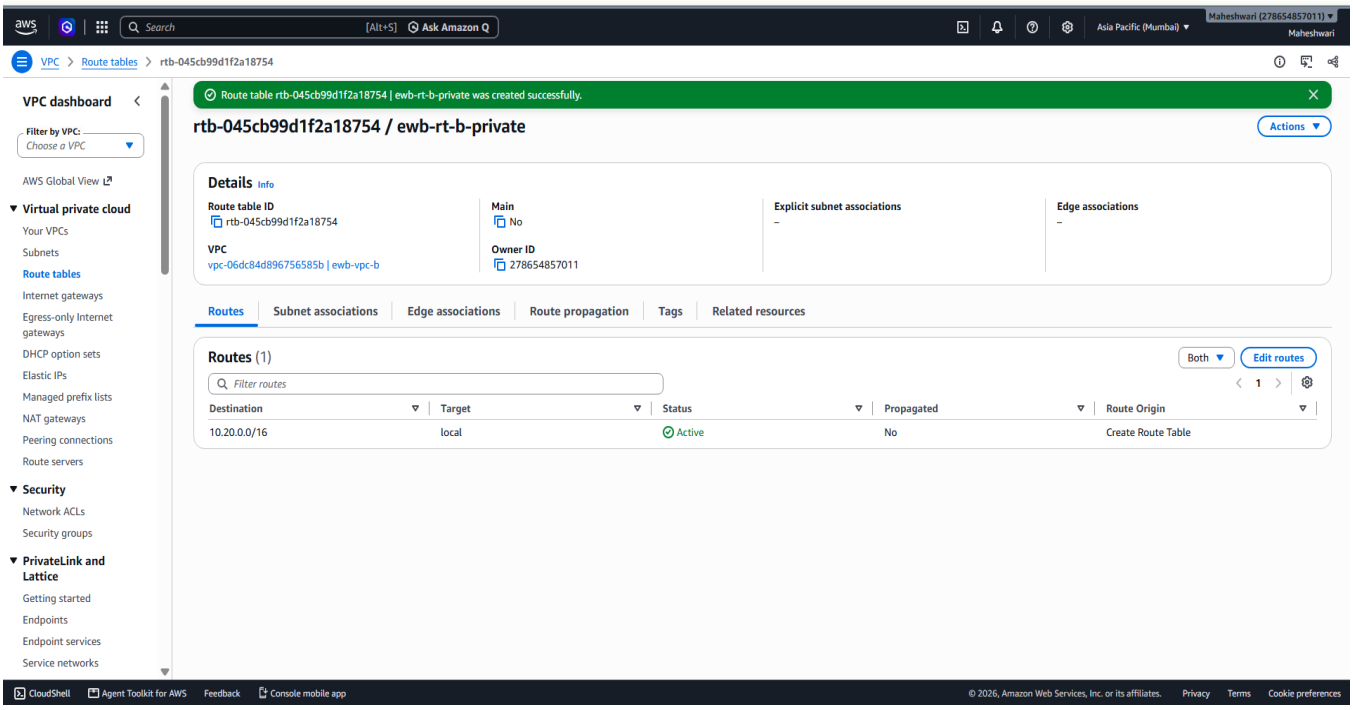


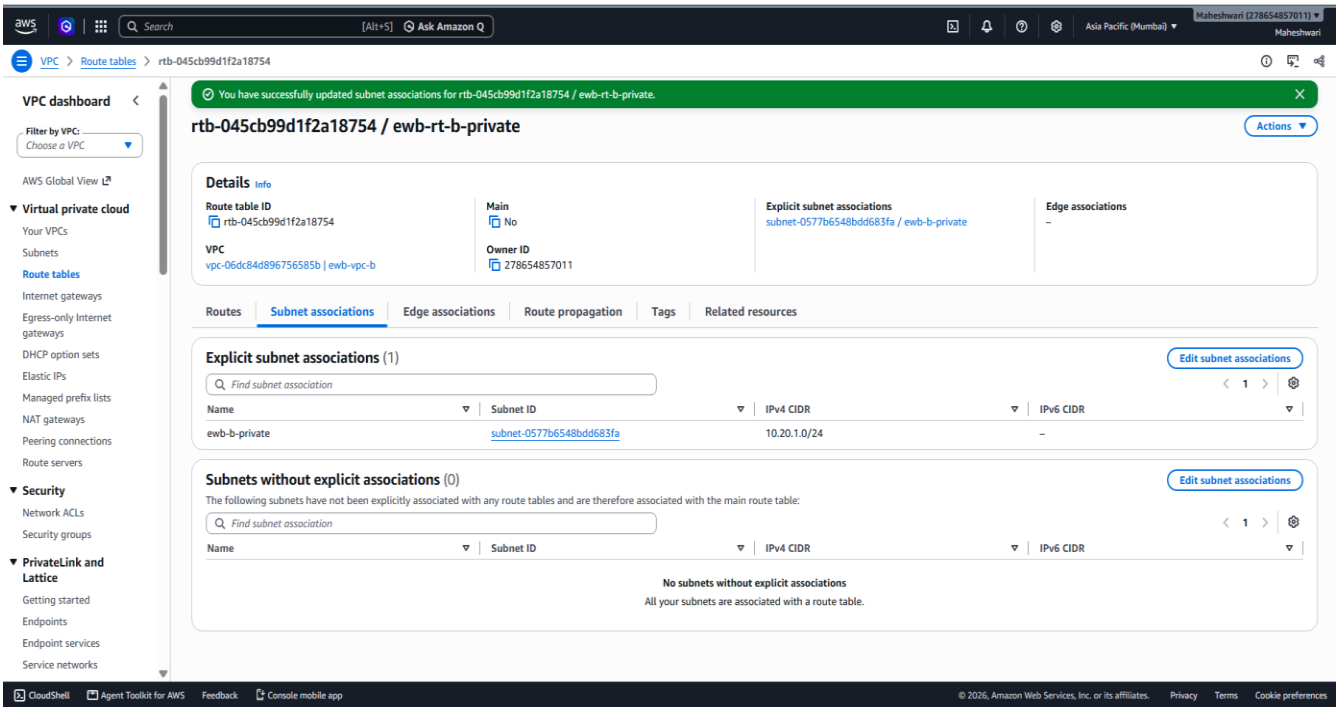


6c: Private route table in VPC B

Private route table in VPC B

ewb-rt-b-private has only the local route (10.20.0.0/16) and is associated with ewb-b-private.

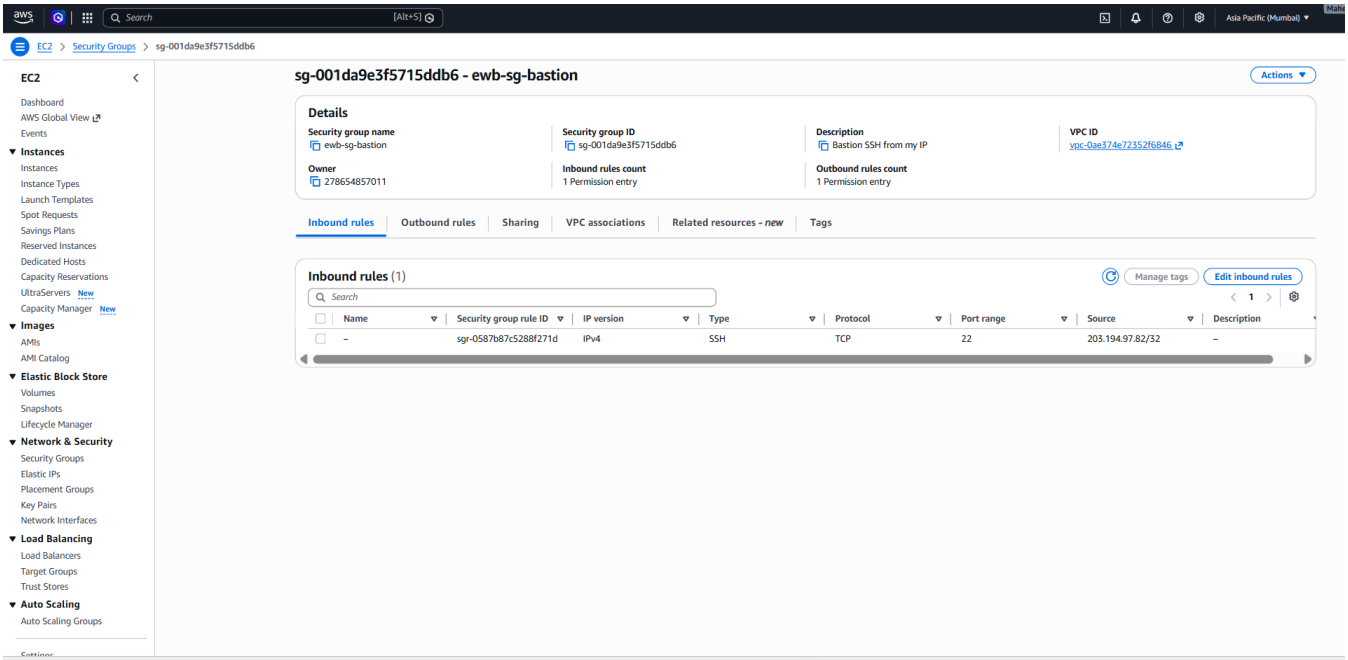




Step 7: Security groups:- Each security group allows only what is needed.

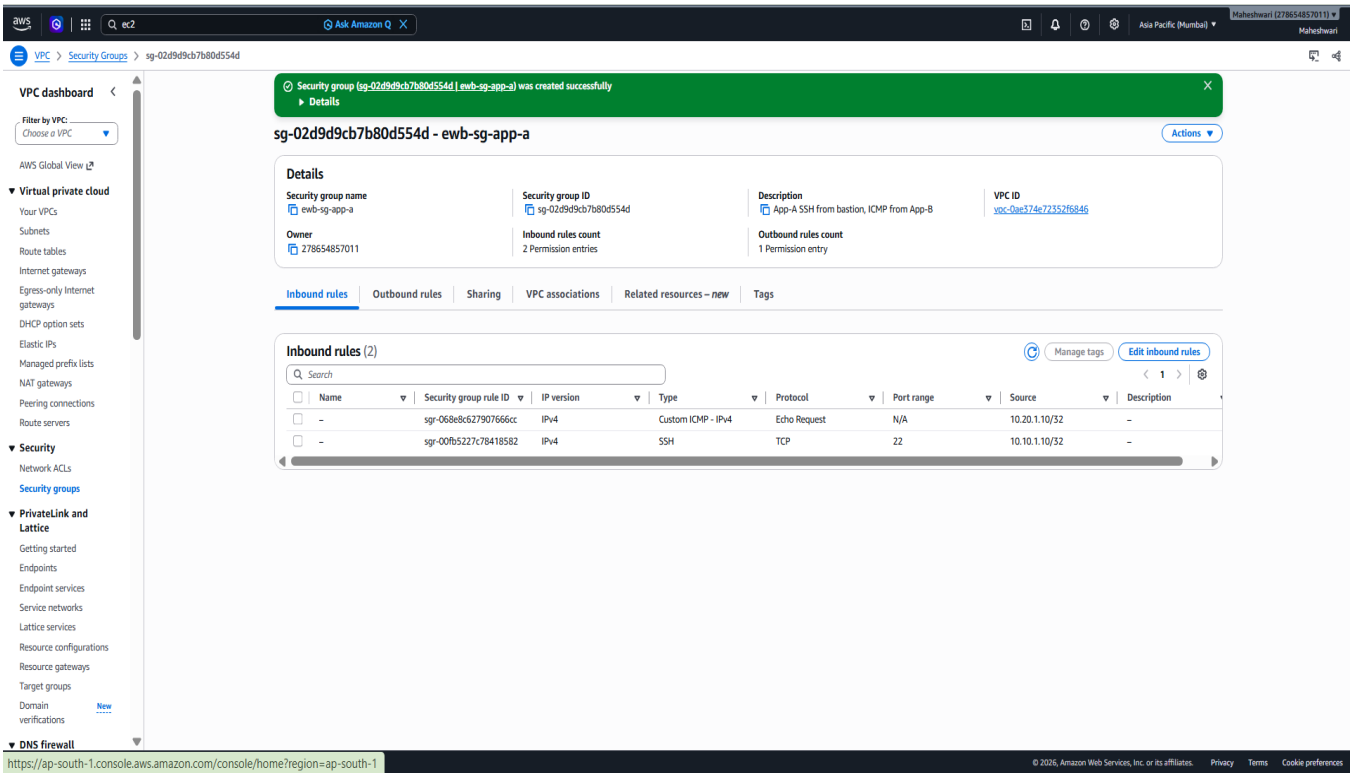
SG 1: ewb-sg-bastion (VPC A)

Allows SSH (port 22) only from my own IP address (/32).



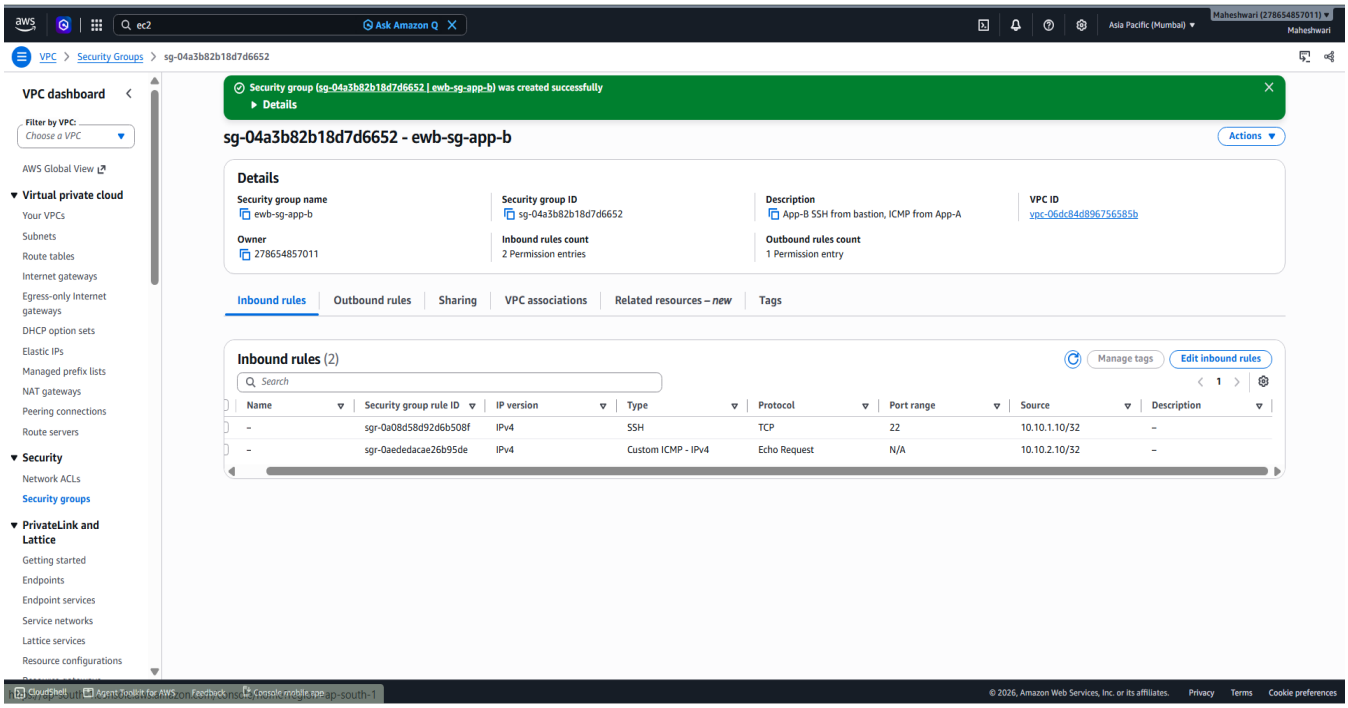
SG 2: ewb-sg-app-a (VPC A)

Allows SSH from the bastion (10.10.1.10/32) and ICMP echo request from App-B (10.20.1.10/32).



SG 3: ewb-sg-app-b (VPC B)

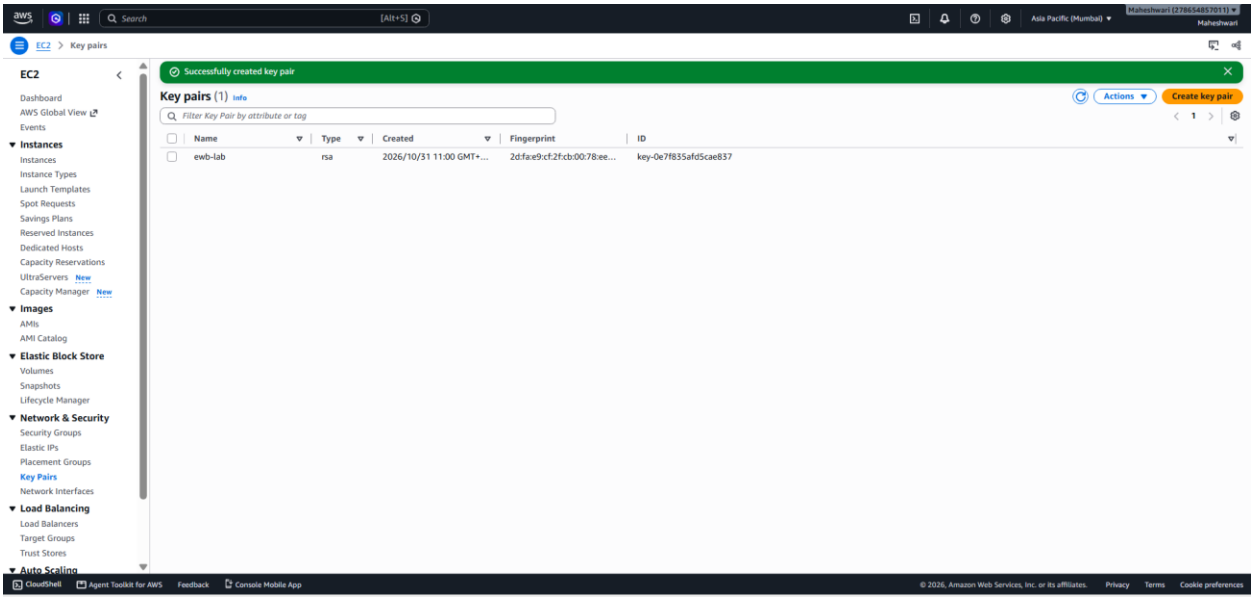
Allows SSH from the bastion (10.10.1.10/32) and ICMP echo request from App-A (10.10.2.10/32).



Step 8: EC2 instances

8A: Key pair

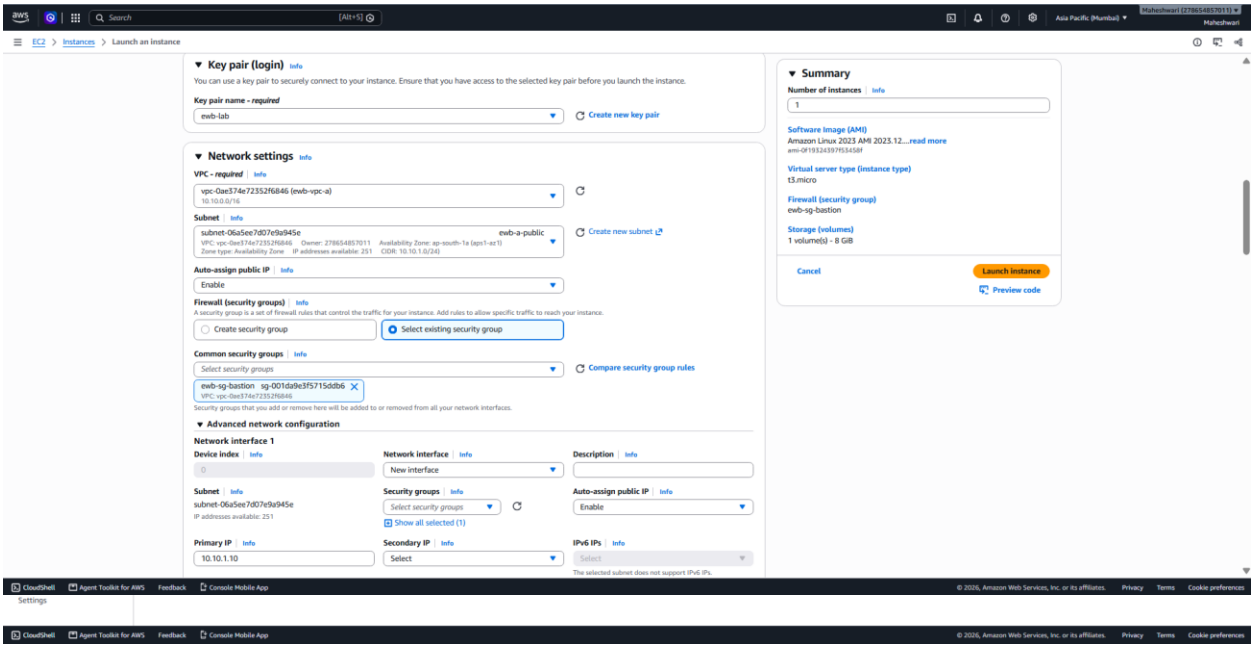
Created the key pair ewb-lab (RSA, .pem). The private key file is kept only on my laptop and is not shared

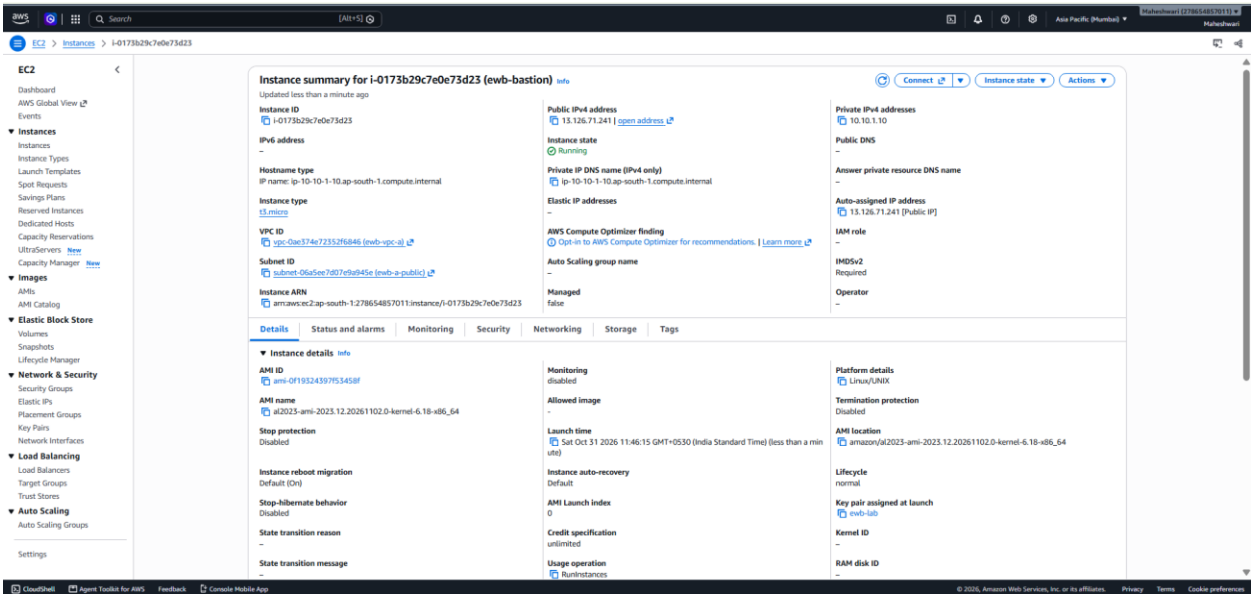


Step 8A: key pair ewb-lab

8B: ewb-bastion

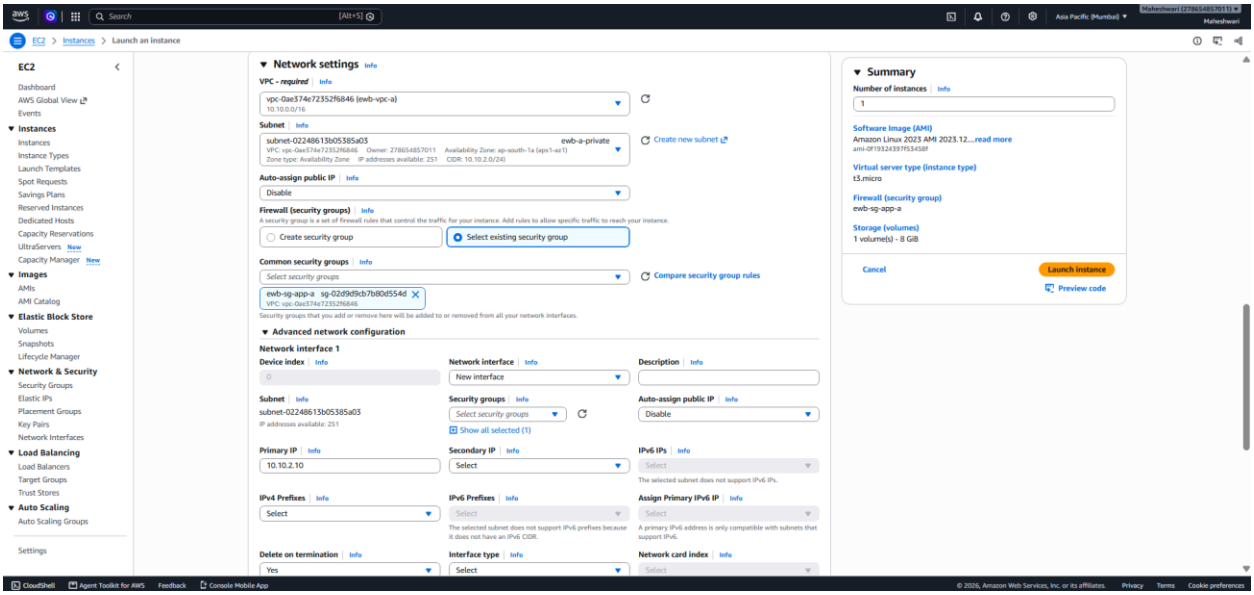
t3.micro, Amazon Linux 2023, ewb-a-public subnet, public IP enabled, private IP 10.10.1.10, security group ewb-sg-bastion.

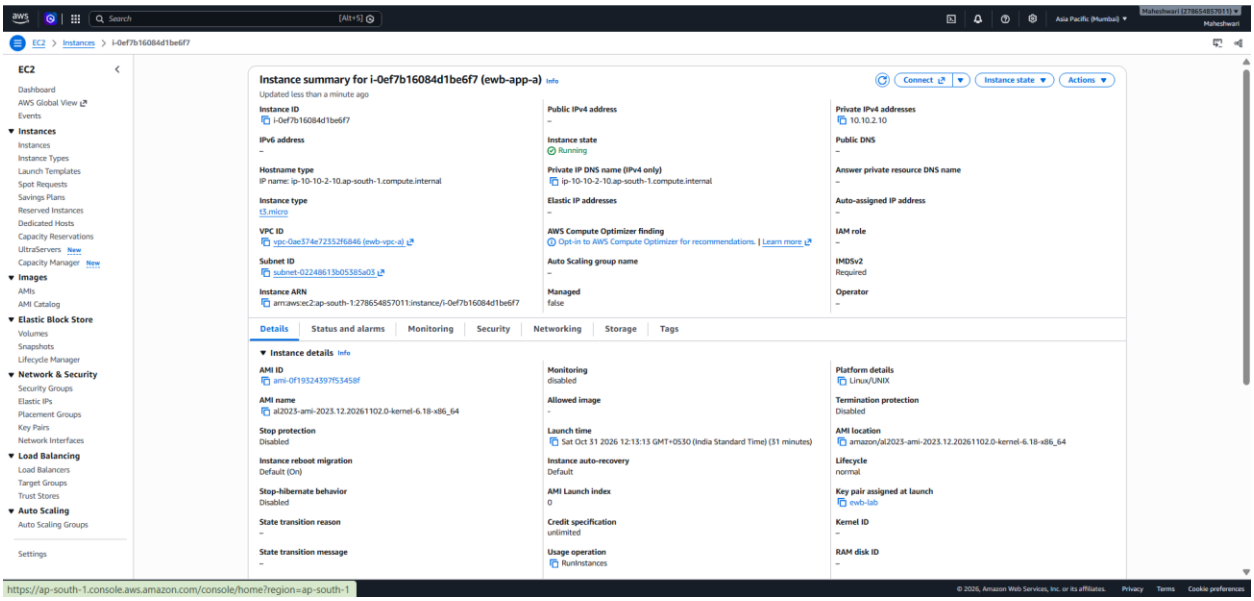




8C: ewb-app-a

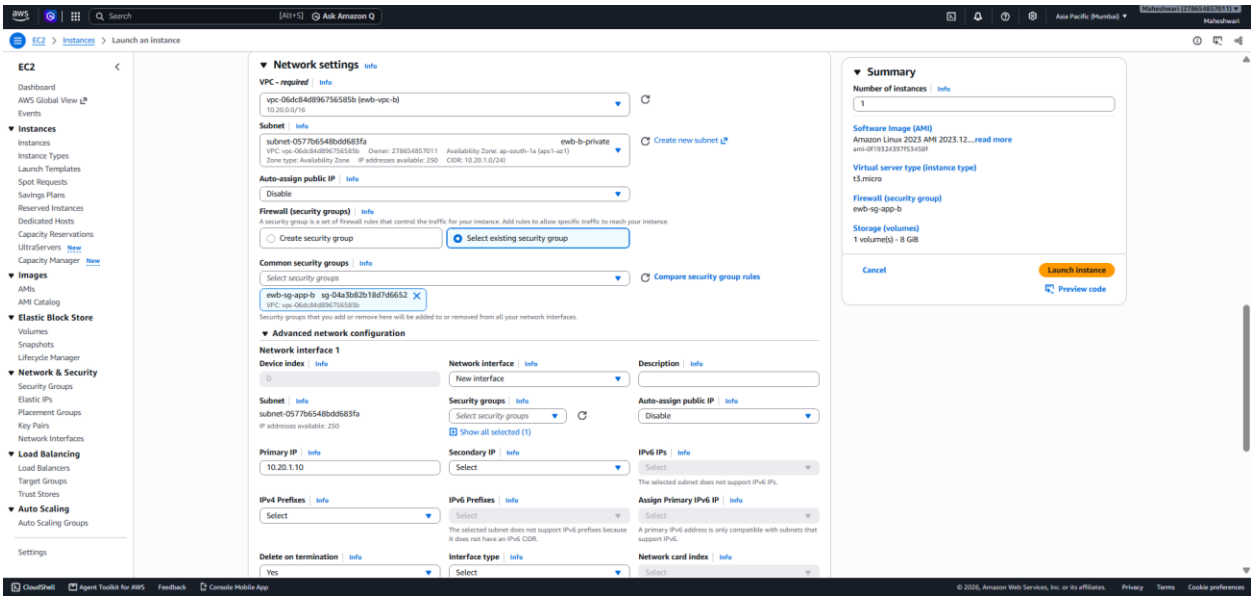
ewb-a-private subnet, no public IP, private IP 10.10.2.10, security group ewb-sg-app-a.





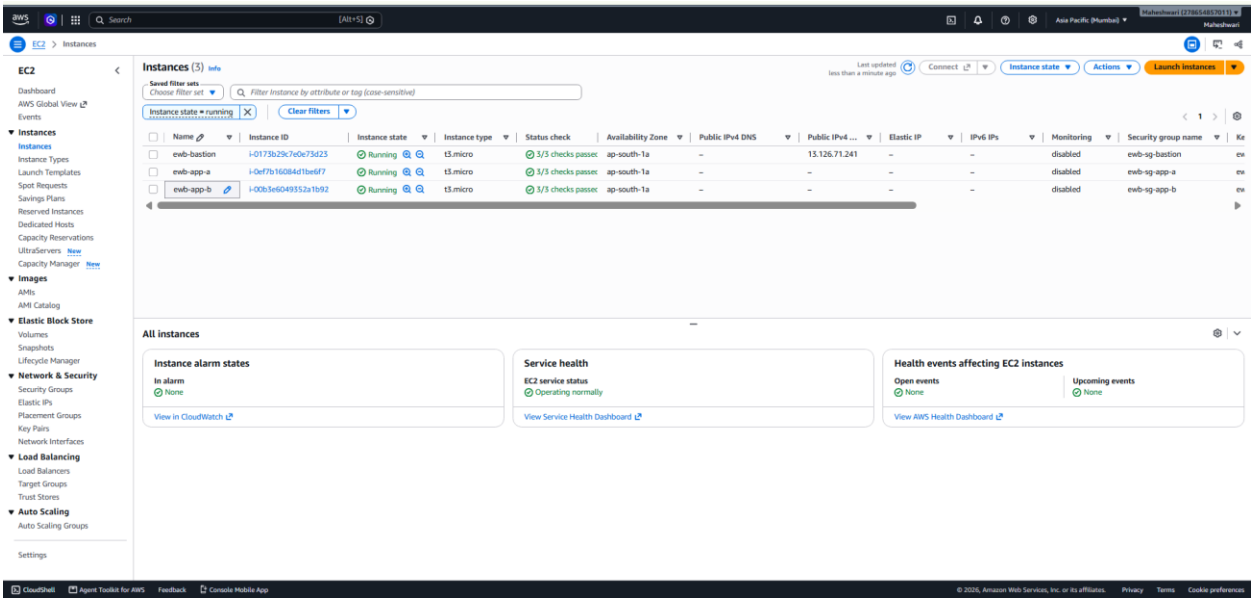
8D: ewb-app-b

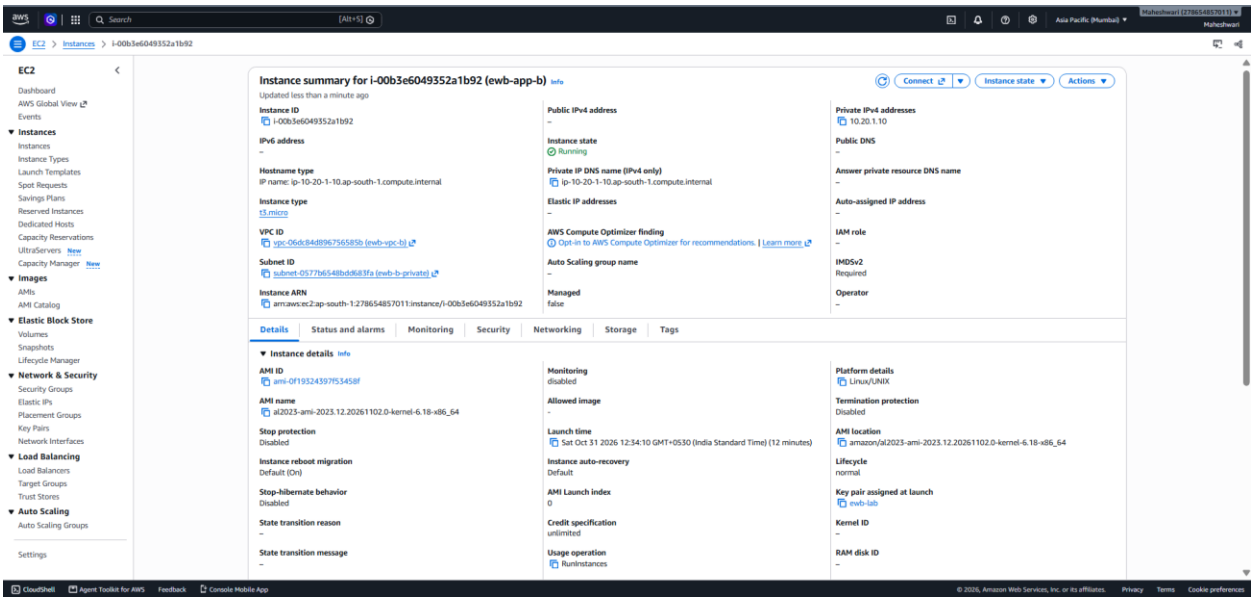
VPC B, ewb-b-private subnet, no public IP, private IP 10.20.1.10, security group ewb-sg-app-b.



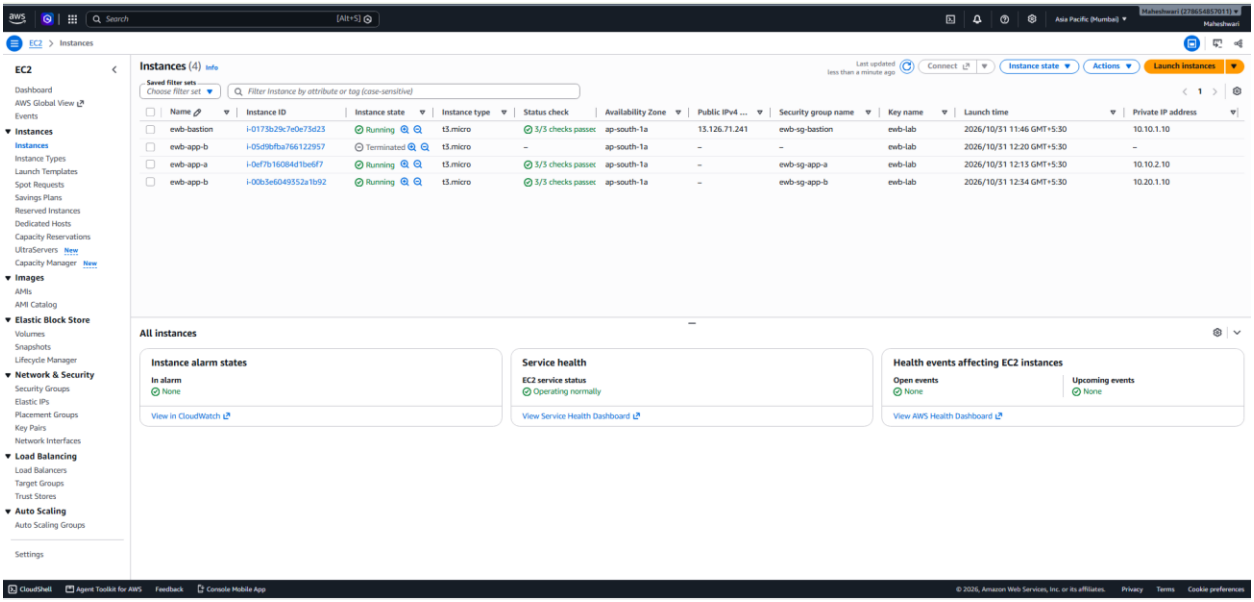
8E: All instances running

All three instances are running and their status checks passed. Only ewb-bastion has a public IPv4 address.



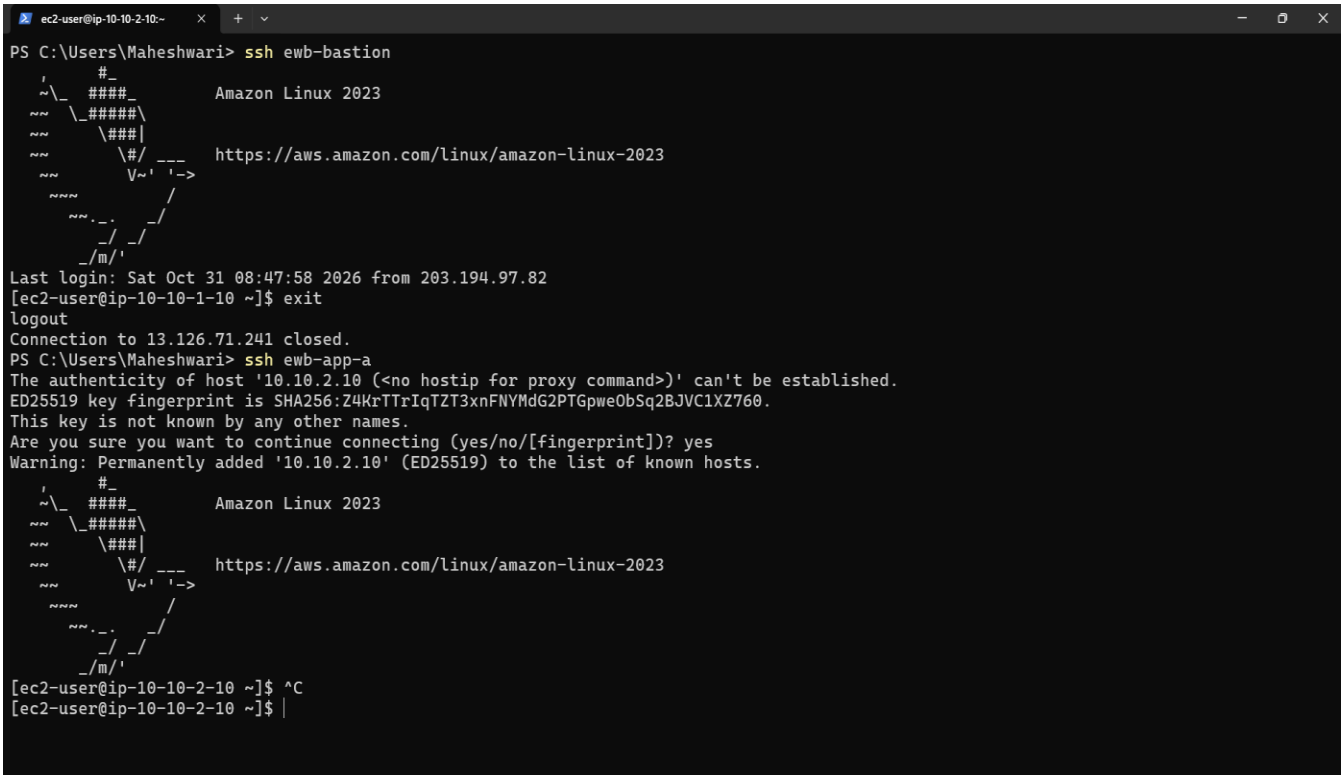


Finally, all instances running successfully.



Step 9: SSH to Bastion and App-A (before peering):-

Connected from the laptop to the bastion host (`ewb-bastion`, public IP), then to the private instance `ewb-app-a` through the bastion using `ProxyJump`. App-A has no public IP, so it can only be reached through the bastion.



Step 10: Baseline test (before peering)

From App-A (`10.10.2.10`), pinged App-B (`10.20.1.10`) in VPC B. No replies (100% packet loss), because the two VPCs are not connected yet.

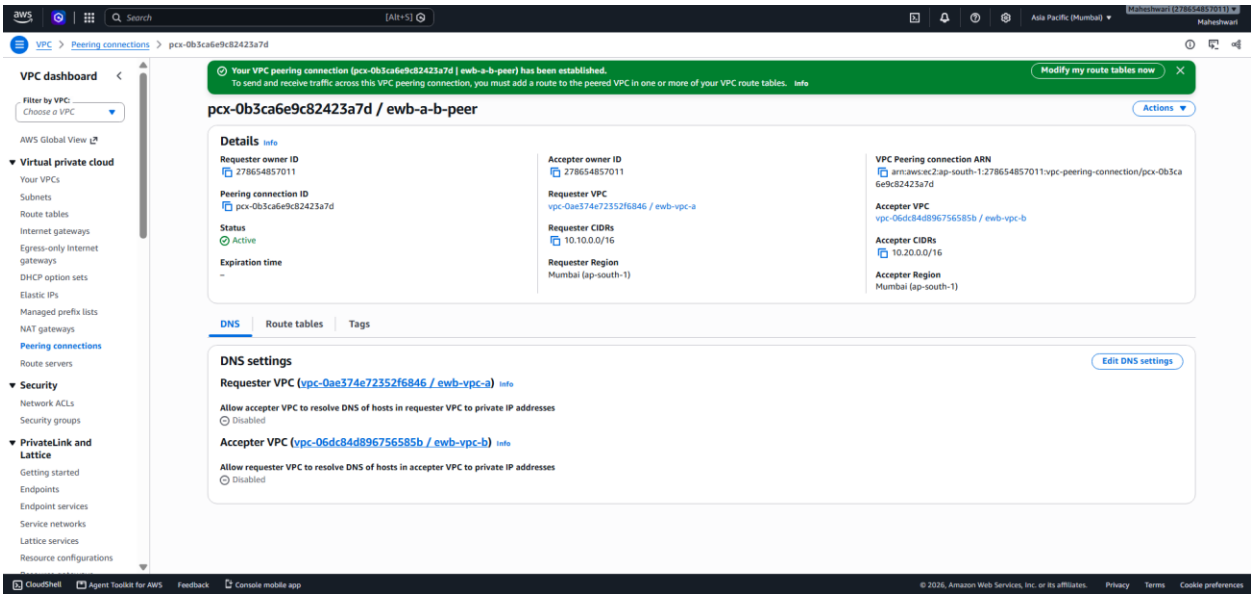
```
[ec2-user@ip-10-10-2-10 ~]$ hostname
hostname -I
ping -c 4 10.20.1.10
ip-10-10-2-10.ap-south-1.compute.internal
10.10.2.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.

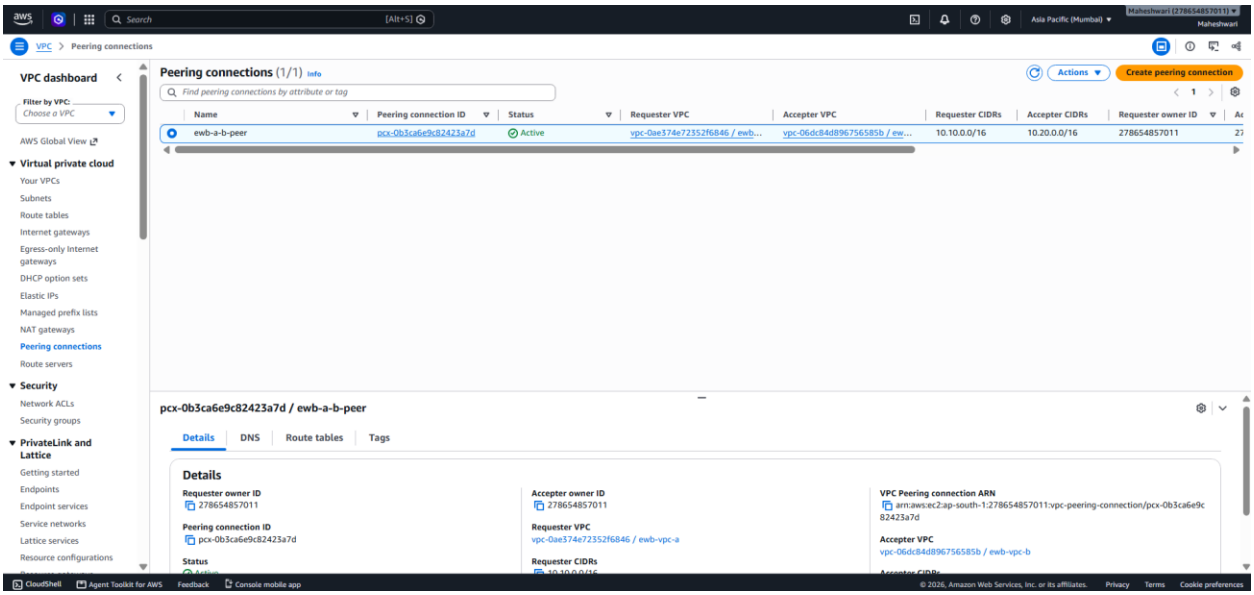
--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3104ms

[ec2-user@ip-10-10-2-10 ~]$
```

Step 11: VPC Peering (AWS Conso Step 11: VPC peering

Created the peering connection ewb-a-b-peer (pcx-0b3ca6e9c82423a7d) with VPC A as requester and VPC B as accepter, in the same account and Region. The request was accepted and the status is Active. le, in the browser)



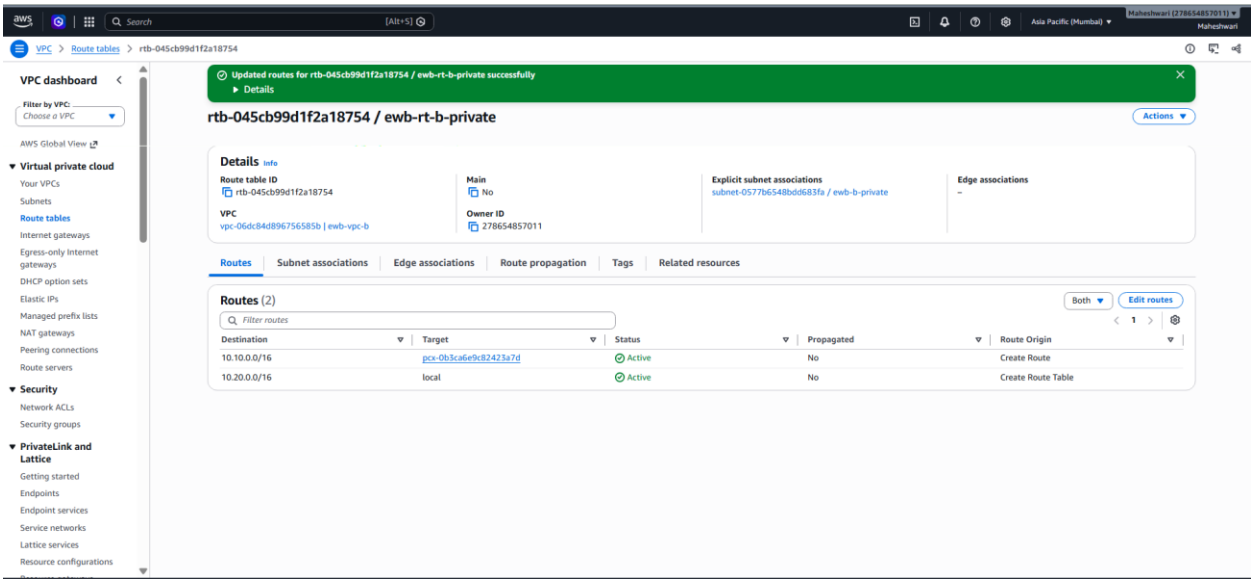


Step 12: Routes for the peering

A peering connection does not carry traffic until each side has a route to the other VPC.

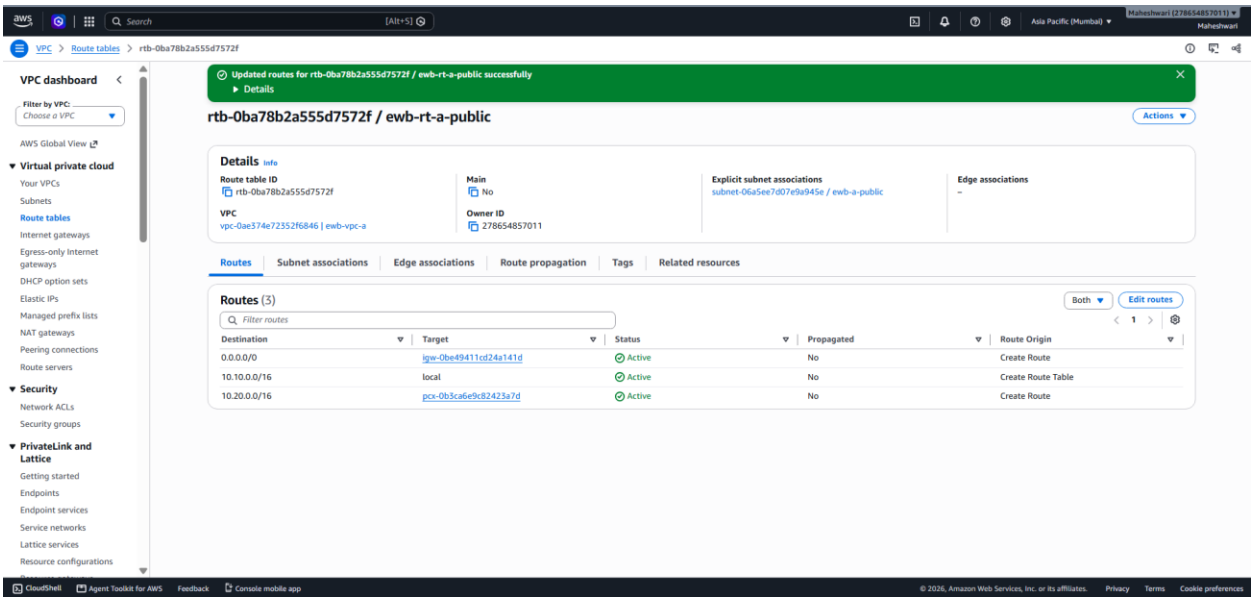
12a: ewb-rt-a-private

10.20.0.0/16 goes to the peering connection (forward path from App-A).

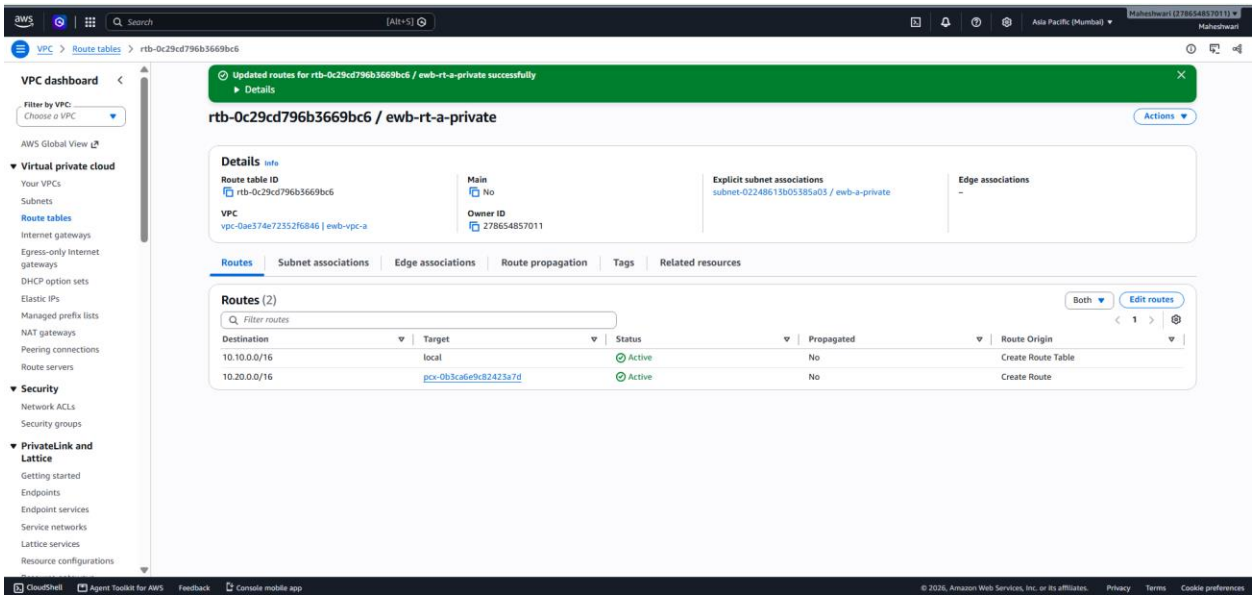


12b: ewb-rt-b-private

10.10.0.0/16 goes to the peering connection (return path from App-B).



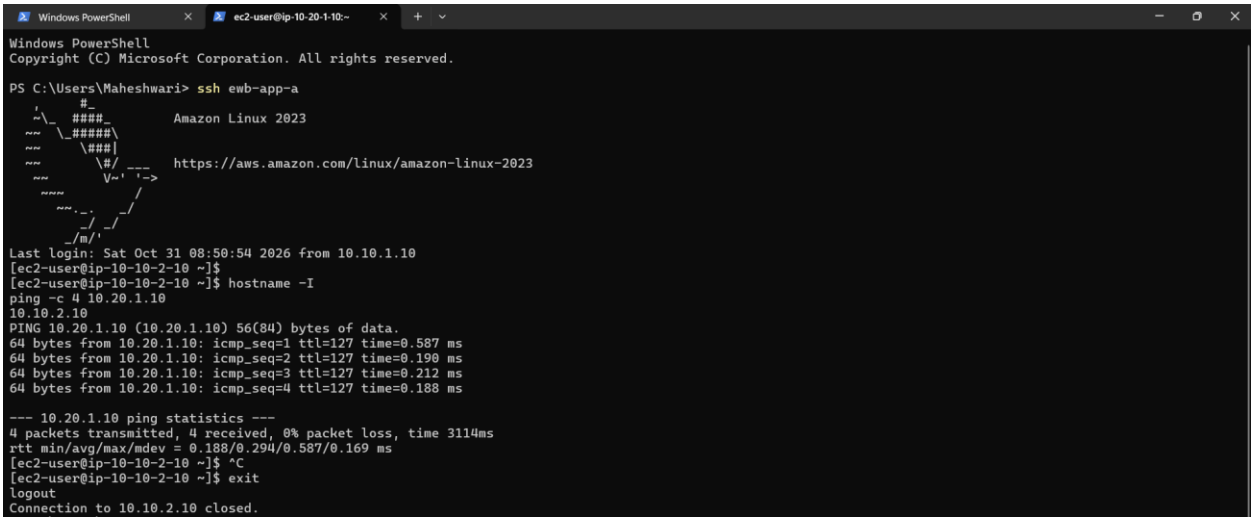
12c: ewb-rt-a-public



Step 13:

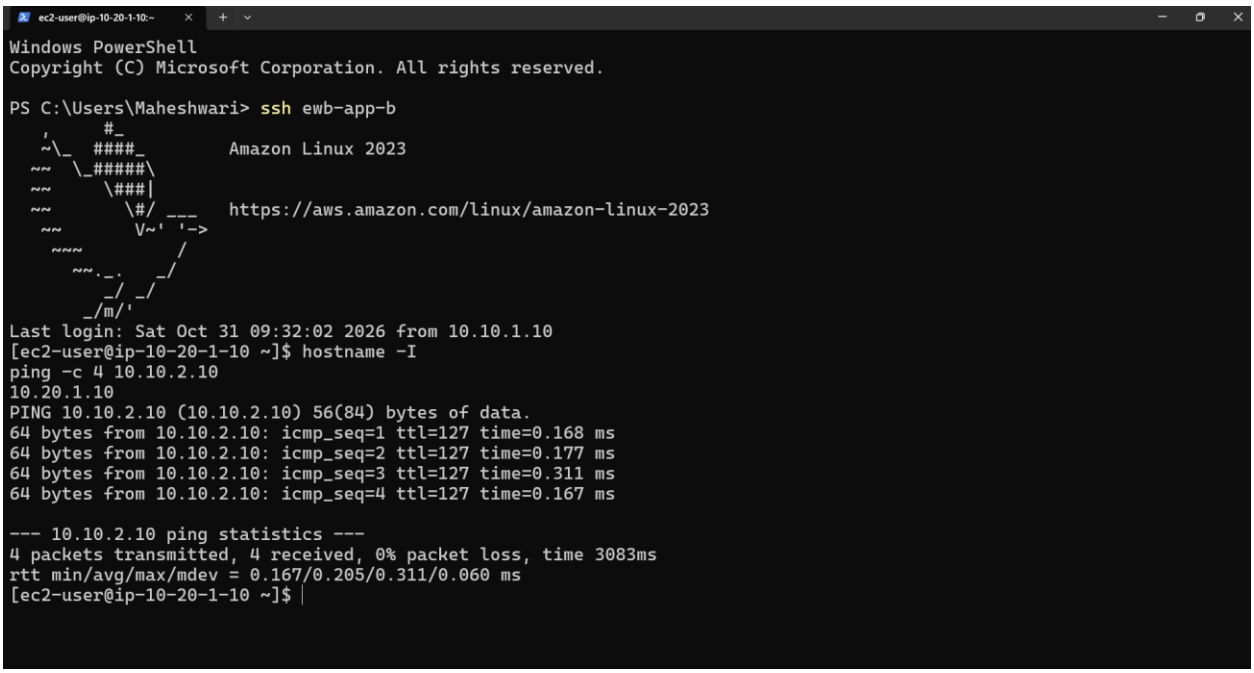
13a: Forward ping (App-A to App-B)

From ewb-app-a (10.10.2.10), ping to ewb-app-b (10.20.1.10) succeeded: 4 packets transmitted, 4 received, 0% packet loss.



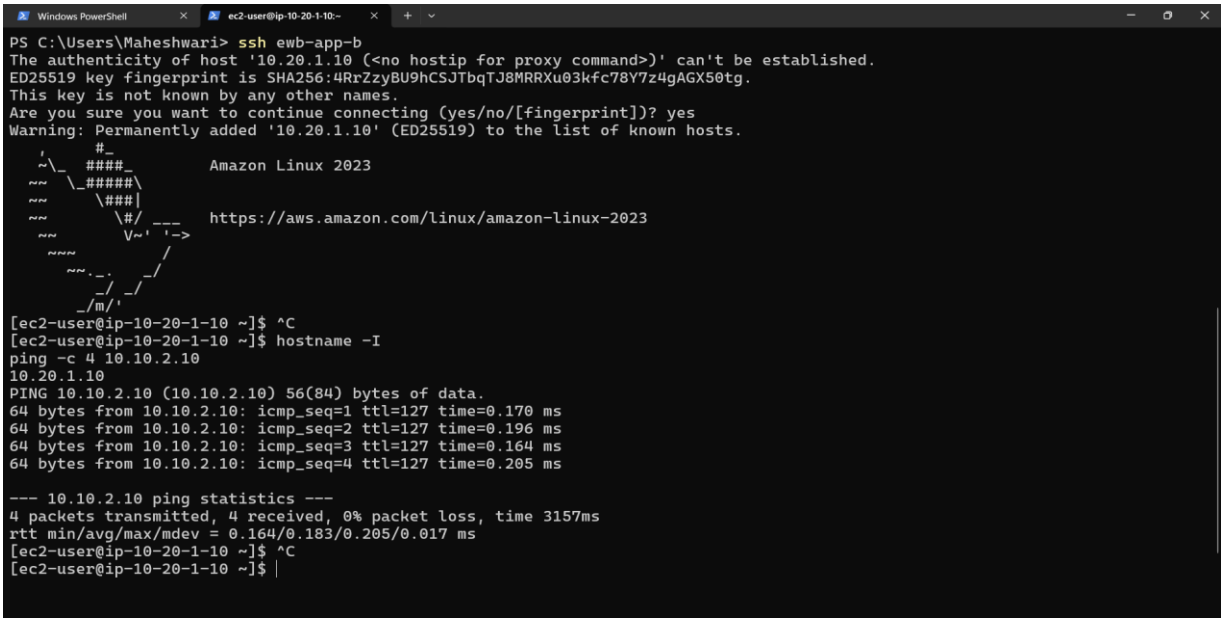
13b: Reverse ping (App-B to App-A)

From ewb-app-b (10.20.1.10), ping to ewb-app-a (10.10.2.10) succeeded: 4 packets transmitted, 4 received, 0% packet loss.



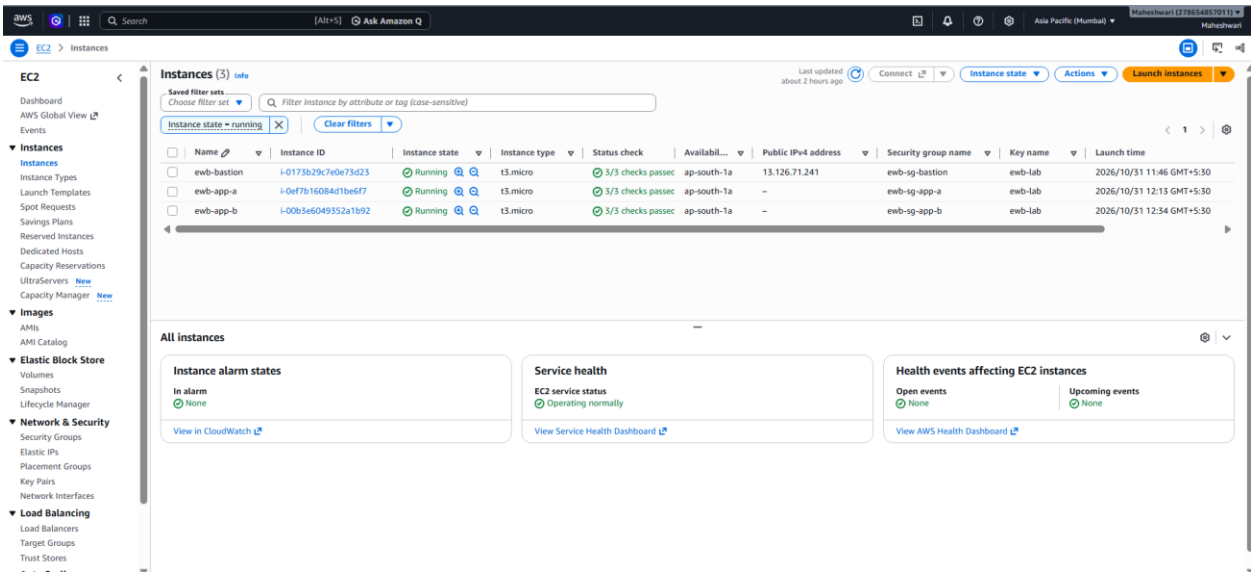
13c: Private SSH to App-B through the bastion:-

ssh ewb-app-b from the laptop reached the private instance in VPC B (prompt ip-10-20-1-10) using ProxyJump through ewb-bastion.



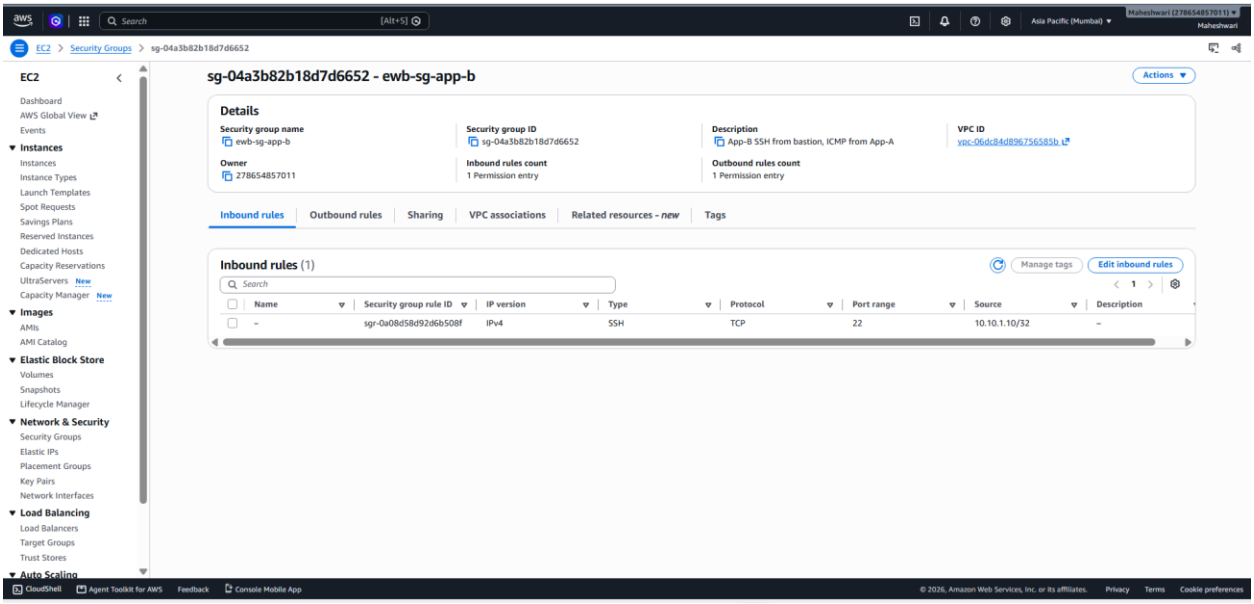
13d: Isolation (no public IPv4):-

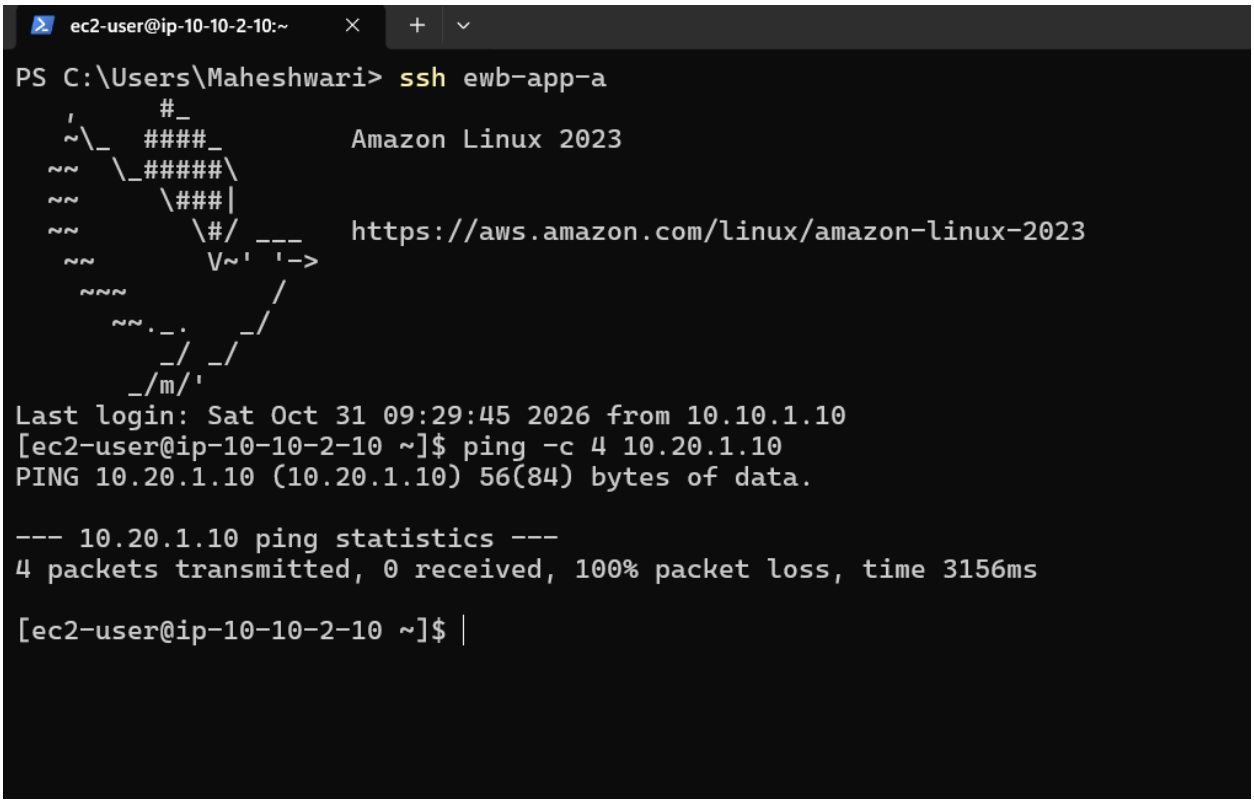
ewb-app-a and ewb-app-b have no public IPv4 address. Only ewb-bastion has one, so the private servers are reachable only through the bastion.



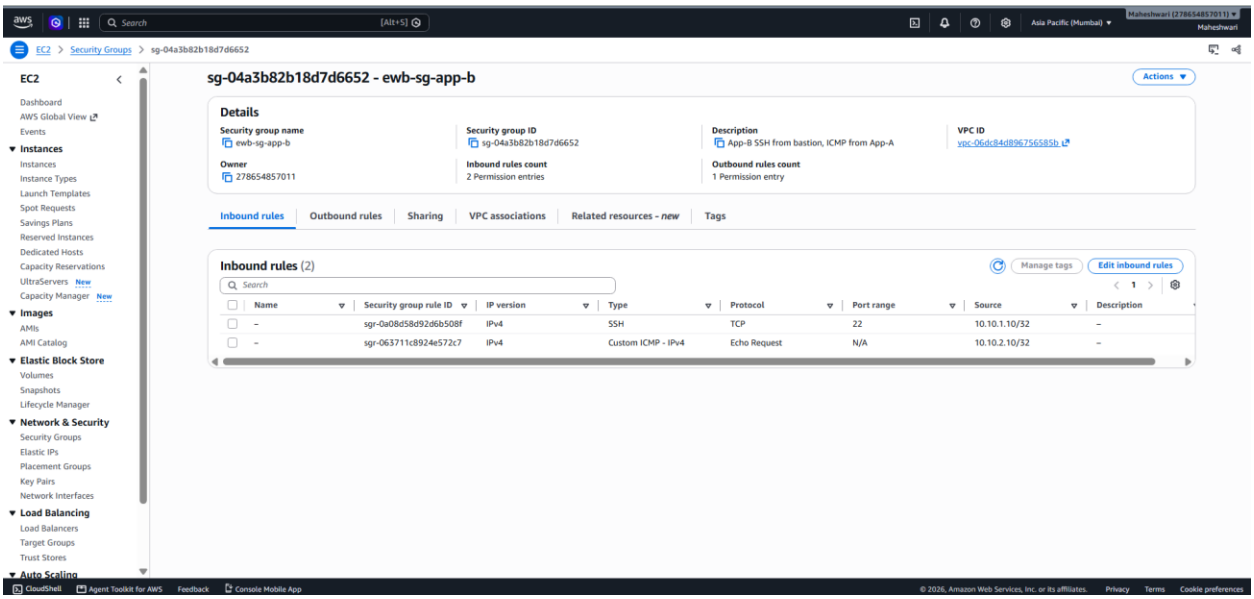
Step 14: Controlled failure

14a: Delete the rule, ping fails





14b: Add the rule back, ping succeeds




```
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.160 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.187 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.190 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.183 ms

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3098ms
rtt min/avg/max/mdev = 0.160/0.180/0.190/0.011 ms
[ec2-user@ip-10-10-2-10 ~]$
```